

Algorithmic Pseudocode Sample 5

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
1: data structure LSH
2: members
3:    $d, n \in \mathbb{N}_+$  ▷  $d$  is dimension,  $n$  is number of data points
4:    $K, L \in \mathbb{N}_+$  ▷  $K$  is amplification factor,  $L$  is number of repetition for hashing
5:    $p_{\text{near}}, p_{\text{far}} \in (0, 1)$  ▷ Collision probability
6:   For  $l \in [L]$ ,  $\mathcal{T}_l := [n]$  ▷ Hashtable recording data points hashed by  $\mathcal{H}_l$ 
7:    $\mathcal{R} := [n]$  ▷ retrieved points
8:    $\mathcal{H} := \{f \in \mathcal{H} : \mathbb{R}^d \rightarrow [M]\}$  ▷  $M$  is number of buckets for hashing family  $\mathcal{H}$ 
9:   For  $l \in [L]$ ,  $\mathcal{H}_l \in \mathcal{H}^K$  ▷ Family of amplified hash functions with at most  $M^K$  non-empty buckets
10:   For  $b \in [M^K]$ ,  $\mathcal{S}_b := \text{AVL tree}$  ▷ Use AVL tree to store points in bucket
11: end members
12:
13: public
14: procedure INITIALIZE( $\{x_i\}_{i \in [n]} \subset \mathbb{R}^d, k, L \in \mathbb{N}_+$ )
15:   CHOOSEHASHFUNC( $k, L$ )
16:   CONSTRUCTHASHTABLE( $\{x_i\}_{i \in [n]}$ )
17: end procedure
18:
19: procedure RECOVER( $q \in \mathbb{R}^d$ )
20:    $\mathcal{R} \leftarrow 0$ 
21:   for  $l \in [L]$  do
22:      $\mathcal{R} \leftarrow \mathcal{R} \cup \mathcal{T}_l.\text{RETRIEVE}(\mathcal{H}_l(q))$  ▷ Find the bucket  $\mathcal{H}_l(q)$  in  $\mathcal{T}_l$  and retrieve all points
23:   end for
24: end procedure
25:
26: procedure UPDATEHASHTABLE( $z \in \mathbb{R}^d, i \in [n]$ )
27:   for  $l \in [L]$  do
28:      $\mathcal{H}_l(z).\text{INSERT}(z)$  ▷  $\mathcal{H}_l(z)$  denotes the bucket that  $z$  is mapped to
29:      $\mathcal{H}_l(x_i).\text{DELETE}(x_i)$ 
30:   end for
31: end procedure
32: end data structure
```
